# W05 Pre-lab Quiz: SPI, I2C, and UART

Started: Feb 4 at 10:18am

# Quiz Instructions

Week 05

Pre-lab Quiz

## Instructions

Review the following videos, and all the study materials in the module then answer the questions.

- **Protocols:  UART, I2C, SPI (https://www.youtube.com/watch?v=lyGwvGzrqp8&ab_channel=Electronoobs)**
- **Understanding I2C (https://www.youtube.com/watch?v=CAvawEcxoPU)**
- **Understanding SPI (https://www.youtube.com/watch?v=0nVNwozXslc)**

Question 1 0.5 pts

Which of the following statements is true?

- ○ I2C communication typically exceeds 1 MHz in standard applications
- ○ SPI is limited to half-duplex communication only
- ○ UART and SPI interfaces are always full-duplex
- ○ UARTs can operate at any speed as long as the clock is configured
- ◉ SPI can be configured to operate at speeds of 30 MHz or higher

Question 2 0.5 pts

Which of the following uses a parity bit to check the integrity of transmitted data?

- ○ None of the above
- ○ I2C
- ○ SPI
- ○ All of the above
- ◉ UART

Question 3 0.5 pts

Which of the following can have the fastest transfer speed?

- ○ UART with flow control
- ○ UART
- ○ I2C with clock stretching
- ○ I2C
- ◉ SPI

Question 4 0.5 pts

Looking at the datasheet for the STM32 (in the module), what is the **minimum output high voltage** for a normal CMOS I/O pin with a loading of 8mA and input voltage (VDDIOX) of 3.3V?

- ○ 1.8V
- ○ 0.4V
- ○ 2.4V
- ◉ 2.9V
- ○ 3.3V

Question 5 0.5 pts

Which of the following communication interfaces requires the master to send an address before transmitting data, and only the slave that recognizes the address will respond?

- ○ SPI
- ○ UART with parity
- ○ UART
- ○ SPI with multiple chip selects
- ◉ I2C

Question 6 0.5 pts

Match each communication protocol to a characteristic feature of that protocol:

| | |
|---|---|
| I2C | Uses a passive signaling sys |
| SPI | Requires mutually-exclusive |
| UART | Typically connects only two d |

Question 7 1 pts

Why does I2C not require extra signals to select the appropriate slave for a transaction?

- ◉ Because each slave device has a unique address, and the master selects the slave by sending its address over the bus
- ○ Because the master and slave are pre-configured to communicate at specific times
- ○ Because the slave devices take turns responding in a round-robin fashion
- ○ Because the master uses a dedicated chip select line for each slave
- ○ Because I2C uses a broadcast mode to communicate with all slaves at once

Question 8 1 pts

What does the TX FIFO empty event indicate?

- ○ The receive FIFO is full and must be read.
- ○ The UART peripheral has been disabled.
- ○ The baud rate has changed and needs to be reconfigured.
- ◉ The last byte has been fully transmitted on the wire.
- ○ The transmit FIFO has space available for new data to be written.

Question 9 1 pts

What four SPI settings are predefined and must be known in advance for both the master and the slave?

- ○ Address, data width, clock skew, and sync pulse
- ○ Clock speed, parity, stop bits, and data length
- ○ Start bit, stop bit, parity, and baud rate
- ◉ Clock polarity, clock phase, packet length, and bit order
- ○ Baud rate, voltage level, data rate, and parity

Question 10 1 pts

How many master devices can be present on an I2C bus at one time?

- ○ One master per slave device
- ○ Unlimited masters can transmit simultaneously
- ○ Multiple masters can be present, but only one can control the bus at a time
- ○ Only one master is allowed on the bus
- ◉ Two masters are allowed, but they must alternate

Question 11 1 pts

In UART communication, how are symbols synchronized?

- ○ By synchronizing both devices to a global system clock.
- ◉ By using start and stop bits to explicitly frame each symbol and align the receiver's clock.
- ○ By embedding a clock signal within the data stream.
- ○ By sharing a clock line between the transmitter and receiver.
- ○ By using edge detection on the data line to infer timing.

Question 12 1 pts

What does the UART_SR register contain?

- ◉ Flags indicating the status of UART operations, such as transmit complete, receive buffer not empty, and error conditions.
- ○ Control bits to enable or disable the UART peripheral.
- ○ The actual data being transmitted or received.
- ○ Configuration bits for setting baud rate, parity, and stop bits.
- ○ Memory-mapped addresses for DMA transfers.

Question 13 1 pts

In serial communications, is symbol synchronization implicit or explicit?

- ○ Implicit, because synchronization is handled entirely by hardware.
- ○ Implicit, due to the use of fixed baud rates.
- ○ Implicit, since the receiver guesses the timing based on signal transitions.
- ○ Explicit, only asynchronous protocols use start and stop bits.
- ◉ Explicit, as it requires framing or control over the clock signal to align symbol timing.

Quiz saved at 10:36am Submit Quiz

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